

Recommendation ideas - Question 1:

Partner actors

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Schema imdb_ijs, MySQL 8.0.40. Every figure is the output of a query that ran. Where a judgement is mine rather than the database's, it is marked as such.

Headline: precision is 84% or 8% on the same sample, depending on the definition. As a detector of performers who repeatedly appear together the filter is accurate; as a source of film acting partnerships it is not, because the pairs it finds are overwhelmingly bands, reality-TV casts and TV duos. Jaccard above 0.5 is only reachable by people with very small filmographies, which is also why recall on famous film duos is zero.

1.1 Find pairs of actors that played together in at least 3 movies

```
create table actor_movies as
select distinct actor_id, movie_id
from imdb_ijs.roles;

create table actor_pairs as
select a.actor_id as actor_1, b.actor_id as actor_2, count(*) as common_movies
from actor_movies as a
join actor_movies as b
  on a.movie_id = b.movie_id and a.actor_id < b.actor_id
group by a.actor_id, b.actor_id
having count(*) >= 3;
```

distinct is needed because roles holds a row per role: an actor credited twice in one film would otherwise count as two shared movies with every co-star.

$a.actor_id < b.actor_id$ handles reflexiveness and symmetry in one condition.

Answer: 1,100,132 pairs.

1.2 Add the condition that the Jaccard measure is above 0.5. What does this condition contribute?

```
where p.common_movies / (c1.movies + c2.movies - p.common_movies) > 0.5
```

Answer: 8,938 pairs remain, a 99.19% reduction.

In principle the two conditions fail on opposite cases. Three shared films is noise for two actors with 200 credits each, and Jaccard catches that by asking what portion of the combined work was done together.

In practice the condition is far too severe, and 1.3 and 1.4 show what it actually selects for. To exceed 0.5 a pair must share more than half of their combined distinct filmography, which is only reachable when both filmographies are small.

A note on the run: CREATE TABLE ... SELECT emitted 2,400 notes of type 1265, data truncated. MySQL infers a DECIMAL type for the stored jaccard column, so the value is rounded on storage. Row selection is unaffected, because the WHERE clause evaluates the full-precision expression before storage. Nothing was lost.

1.3 Choose 10 pairs that were retrieved and decide if they are indeed partners. What is the precision?

The sample is drawn with order by rand() from the 8,938 retrieved pairs. Ordering by shared movies instead would be an over-estimate, since the pairs with the most shared films are the most likely to be genuine.

I judged each pair from the titles of the films they actually share, pulled from the database. A pair counts as a partnership only if both are film actors who worked together repeatedly.

#	Pair	Jaccard	What the shared films are	Partners?
1	Jimmy Destri & Chris Stein	0.588	Blondie concert films and music documentaries	No
2	Aaron Bailey & Beth Anthony	0.833	The Real World, same season plus reunion specials	No
3	George Figgs & Susan Walsh	0.571	Pink Flamingos, Female Trouble - John Waters troupe	No
4	Herman Rosen & Jacob Zanger	0.667	Yiddish cinema, 1939 to 1940	Yes
5	Jane Nejt & Rie Pedersen	0.800	Robinson ekspeditionen, Danish Survivor	No
6	João Barone & Herbert Vianna	1.000	Os Paralamas concert films	No
7	Alyson Kiperman & Jessica Rey	0.750	Power Rangers Wild Force episodes	No
8	Maurice Dembsky & Ed Livingston	0.750	Country-music films, 1966 to 1968	Yes
9	Erik Bragg & Lily Marshall-Fricker	1.000	Bit parts across Garry Marshall films	No
10	Bodo Hauser & Ulrich Kienzle	1.000	German TV news programmes	No

precision = 2 / 10 = 0.2

Ten is a small sample, so I repeated it with 50 random pairs, classified each by what the shared titles show, and scored them under two definitions of partnership. The narrow one counts only film acting partnerships. The broad one counts any two performers who genuinely recur together, whatever the format.

Category	Pairs	Broad	Narrow
Reality TV	12	counts	-
Music act	11	counts	-
TV series	8	counts	-
Film ensemble or troupe	7	counts	-
Genuine film acting partnership	4	counts	counts
Documentary co-subjects, unidentifiable	8	-	-
Total qualifying		42	4

precision, broad definition = 42 / 50 = 0.84

precision, narrow definition = 4 / 50 = 0.08

The definition drives the number, not the method. Under the broad reading the filter is accurate: 84% of what it returns really is two performers who keep appearing together. Under the narrow reading it is nearly useless for the task, because only 8% are the film partnerships a movie recommender wants.

Which reading is right depends on what the partner table is for. Repeated co-appearance is the same behaviour whether it happens in films, in a series or on stage, so counting series and music acts is legitimate. But as the next section shows, no single threshold can serve both: a series with dozens of episodes reaches a high Jaccard easily, while two film stars never can. The populations are not comparable on one scale.

The 8 excluded pairs are documentary co-subjects - real people captured together in archival footage, such as flight directors in Apollo films or the cohort of the Up series - plus two I could not identify. They are subjects appearing together, not performers recurring as an act, which is why they fail both definitions. Counting the identifiable ones would raise the broad figure towards 46 of 50; 42 is the conservative reading.

How the classification was done.

For each sampled pair I pulled the titles and years of the films they actually share, straight from the database, and worked out from those what the pair is. Two people whose shared titles are all concert films of one band were marked as a music act. Two whose shared titles are one season of a reality show plus its reunion specials were marked as reality TV. The same reading identified series, ensembles and documentary subjects.

So the titles are data and the category is my interpretation of them. There is no such field in `imdb_igs`. The reality-TV, music and documentary calls are unambiguous once the titles are visible; the line between an ensemble and a duo is the soft one, and it is the only place where a different reader would plausibly score differently.

This could be partly automated - flagging pairs whose shared titles all contain words like Live or Concert, or all fall in a single year - but on 50 pairs reading the titles was faster and more reliable than any rule I could write.

1.4 Check the recall. Think of 10 pairs of partners and check how many of them were retrieved by the query

A caveat first, because it limits what this measurement can mean. Measuring recall properly requires a random sample from the true population of partners, and no such catalogue exists. Nominating pairs from memory produces a set biased towards famous duos, so the figure below is an estimate over well-known partnerships rather than over all partnerships. That bias cannot be removed without a gold-standard list.

The pairs, chosen before looking at any result:

#	Pair	Shared	Jaccard	Why missed
1	Adam Sandler & Rob Schneider	19	0.218	Jaccard
2	Matt Damon & Ben Affleck	16	0.157	Jaccard
3	Ben Stiller & Owen Wilson	10	0.093	Jaccard
4	Daniel Radcliffe & Rupert Grint	6	0.500	Boundary - excluded by the strict inequality
5	Vince Vaughn & Jon Favreau	5	0.070	Jaccard
6	Jackie Chan & Chris Tucker	5	0.034	Jaccard

recall = 0 / 6 = 0.0

I originally nominated ten. Four of them - Jim Carrey and Jeff Daniels, Christian Bale and Michael Caine, Jonah Hill and Channing Tatum, Dwayne Johnson and Kevin Hart - share one film or none in this snapshot, because their collaborations postdate it. A pair that cannot enter actor_pairs at all cannot test the retrieval rule, so counting them would understate recall for a reason that has nothing to do with the model. They are excluded from the denominator and reported here instead.

All names resolved to actor ids; no pair was missed through a lookup failure. Three needed disambiguation against same-name entries: Rob Schneider, Jackie Chan and Chris Tucker.

Of the six that could be retrieved, five were blocked by the Jaccard condition despite sharing 5 to 19 films. The highest Jaccard among them is 0.218, less than half the threshold. They fail because each actor also has a large independent filmography, and every solo film enlarges the union in the denominator.

The sixth is the boundary case, and it is exact. Radcliffe and Grint share 6 films and have 7 and 11 credits, giving $6 / (7 + 11 - 6) = 0.5000$ precisely. The rule is a strict inequality, so the pair is excluded by nothing at all. Changing $>$ to $\geq$ admits it, at a cost of 2,474 extra pairs across the whole result, and it is the only one of the six affected.

1.5 How can the precision be improved?

The 1.3 results show the errors are not calibration errors, so raising the threshold is not the lever.

Exclude non-film formats.

62% of the sample were reality TV, music acts or TV series, and a further 16% were documentary subjects. Removing them would eliminate the largest error class outright, without touching the ratio at all.

The obstacle is that imdb_igs has no field for this. There is no column saying whether a title is a film, a series episode or a concert recording, so the distinction has to be modelled rather than looked up. Three signals available in the data:

- Title patterns. Concert films and specials carry words such as Live, Concert, Tour or In Concert; series episodes often repeat a parent title with an episode name appended.
- Temporal shape. A reality season plus its reunion specials sits inside one or two years, and a series produces many titles in a tight band. A film partnership is spread across years.
- Cast-overlap shape. Episodes of one series share nearly their whole cast with each other, which distinguishes them from separate productions that happen to reuse two actors.

None of these is exact, and any of them would need to be validated against a hand-labelled sample before being trusted - which is the same procedure used in 1.3. The point is that the fix is a modelling problem rather than a filtering one, and that is why it is worth stating separately from threshold tuning.

Decide whether an ensemble counts.

14% were ensembles, where a whole troupe co-appears and every member pairs with every other. This is a question of definition rather than a defect. A repertory company is a real signal about what kind of film something is, and for a recommender that may be exactly what you want: two films sharing four members of the John Waters troupe resemble each other more than two films sharing one actor. Under the broad definition in 1.3 these seven pairs count.

If the goal specifically is two-person partnerships, they can be separated by capping the number of mutual co-appearances, or by excluding films whose cast overlaps almost entirely with another film in the same set. But that is a choice to make deliberately, not an error to correct.

Deduplicate actor identities.

A catch-all actor id produces spurious high-Jaccard pairs at any threshold. In an earlier run, Roy Rogers paired with an actor whose name appears across his entire 84-film catalogue from 1938 to 1976 - the signature of a mis-merged id rather than a collaborator. Worth noting because that pair clears both conditions decisively and is still wrong: a metric can only be as clean as the entity resolution underneath it.

And a counter-intuitive one: lowering the threshold may raise precision.

Normally precision and the threshold move together. Here they may not. The population above 0.5 is structurally made of small-filmography clusters, so loosening the threshold admits actors with larger careers - the ones who look like film partners. If the target is film partnerships rather than co-appearance clusters, a lower threshold combined with a minimum shared-film count could improve precision and recall at the same time. That is a hypothesis this sample suggests and does not prove.

1.6 How can the recall be raised?

Measured rather than argued. Sweeping the Jaccard cutoff at a fixed 3-film floor, against the six reachable pairs:

Jaccard cutoff	Pairs retrieved	Known pairs recovered	Recall
> 0.50 (original)	8,938	0	0.00
> 0.25	32,330	1	0.17
> 0.20	46,911	2	0.33
> 0.15	75,060	3	0.50
> 0.10	136,998	3	0.50
> 0.05	346,614	5	0.83

Changing the ratio does not help either. The overlap coefficient, shared divided by $\min(\text{movies}_1, \text{movies}_2)$, is strictly more forgiving than Jaccard because it uses the smaller filmography rather than the union. It retrieves 8.9 times as many pairs and recovers the same single pair.

Rule	Pairs retrieved	Recovered of 6	Recall
Jaccard > 0.5	8,938	0	0.00
Jaccard >= 0.5	11,412	1	0.17
Overlap > 0.5	79,111	1	0.17
common_movies >= 5	305,952	6	1.00
Hybrid: common >= 5 and overlap >= 0.15	132,012	5	0.83

The reason is now measurable across the whole result set, not just a sample:

Both actors have at most	Pairs	Share of retrieved
5 films	5,839	65.3%
10 films	8,031	89.9%
20 films	8,624	96.5%
50 films	8,840	98.9%

In 96.5% of retrieved pairs both actors have 20 films or fewer, and only 1.1% contain even one actor with more than 50. The famous duos live at 60 to 145 films each. They are not missing by accident - a large filmography makes Jaccard 0.5 arithmetically unreachable, so they cannot appear at any cutoff near 0.5.

So the answer is that no ratio recovers them. An absolute count does: `common_movies >= 5` retrieves all six reachable pairs, at a cost of 305,952 candidates. The hybrid, `common >= 5` with `overlap >= 0.15`, keeps five of the six and cuts the candidate set by 56.9%. The pair it loses is Vaughn and Favreau at overlap 0.147, missing by three thousandths - which, together with Radcliffe and Grint landing on exactly 0.500, means two of six were decided at a boundary.

1.7 How can partners be used in order to raise the precision of an actor-based recommendation model?

An actor-based recommender suggests film Y to someone who liked film X when the two share an actor. One shared actor is weak evidence: a background credit in both fires the same rule as a lead role in both.

Requiring the shared element to be a partner pair raises the bar. Two films sharing a duo are more likely to be alike in genre, tone and era than two films sharing one person. Fewer recommendations survive and a larger share are right.

This only works if the partner table contains film partnerships. On the current table it would not - recommending a film because it shares two members of the same band is a correct co-appearance and a useless recommendation.

1.8 How can partners be used in order to raise the recall of an actor-based recommendation model?

The same object in the opposite direction. If the viewer liked a film with actor A, the direct rule reaches only films containing A. Extending to films containing A's partners reaches films where A never appears.

The justification is that a partner is a proxy for the same kind of film. This produces candidates the direct rule cannot generate, which is what raising recall means.

1.9 How will raising the precision of the partners model influence the recommendation model?

It depends on which role the partner table plays, and the two roles pull in opposite directions.

As a filter (1.7), higher partner precision is a straight improvement: fewer spurious duos, fewer weakly justified recommendations.

As an expander (1.8), the same tightening produces fewer partners and therefore fewer candidates, so recommendation recall falls.

Concretely: moving from `common_movies >= 5` to the Jaccard rule cuts the partner table from 305,952 to 8,938, retaining 2.9%. As a filter that is an improvement. As an expander it removes 97.1% of the expansion capacity, and the survivors contain none of the six recognisable duos. A component's quality metric does not transfer to the system unchanged; it transfers through the way the component is used.

1.10 How will raising the recall of the partners model influence the recommendation model?

The mirror image, with one asymmetry. As an expander, more partners means more candidates and higher recommendation recall. As a filter, the same growth admits weaker pairs and lowers precision.

The asymmetry is that expansion is multiplicative rather than additive: each partner added brings that partner's entire filmography into the candidate set, not one film. Dropping the cutoff from 0.50 to 0.05 multiplies the partner table by 38.8; `common_movies >= 5` multiplies it by 34.2. A modest gain in partner recall can produce a very large increase in recommendation volume, and the added recommendations come from the pairs that only

just cleared the loosened threshold. Capping the number of partners taken per actor is worth more here than tuning the threshold.

Method

Queries were run against a local MySQL 8.0.40 instance holding `imdb_igs`, using an AI coding assistant to execute them and return the output. Every number is the output of a query that ran. Where a result was surprising, the underlying rows were pulled and inspected before the conclusion was written.

Two things here are judgement rather than data: whether each sampled pair counts as a partnership in 1.3, and which pairs to nominate in 1.4. Both are stated so a reader can disagree without re-deriving the measurements. The samples are stored in the tables `sample10` and `sample50`, so the classifications can be checked pair by pair.